

EDUCATION

Fordham University, Gabelli School of Business*MS, Quantitative Finance*

GPA: 3.75/4.00 (Dean's Scholarship)

- Stochastic Calc, Comp Finance, Adv Credit Risk Modeling, Crypto Algo trading, Market Microstructure

New York, NY

May 2024 – May 2026

Pune Institute of Computer Technology (SPPU)*Bachelor of Engineering, Computer Science*

GPA: 3.53/4.00

- Graduated with honors in Data Science with distinction (4.00/4.00)

Pune, India

August 2018 – August 2022

RESEARCH EXPERIENCE

Research on Fractional Differencing with Prof. Dmitry Udler

September 2025 – March 2026

- Developed finite-sample estimator of market memory ($\hat{d}^{LTM}$) via ACF tail detectability boundaries, addressing failure of standard ARFIMA methods (Extended Whittle collapses for **83.9%** of S&P 500 assets)
- Walk-forward on 484 S&P 500 constituents (2010–2025): outperforms fixed $d = 0.5$ in **70–90%** of assets monthly (Wilcoxon $p < 0.0001$) on residual whitening; dominates $d = 0.5$ on ML forecast correlation in low- $\hat{d}^{LTM}$ regime
- Used $\hat{d}^{LTM}$ regimes to document systematic deepening of return skewness from -0.14 to -0.46 across memory regimes; two working papers drafted ([Market Memory & Over-Differencing](#), [Finite-Sample Long-Memory Estimation](#))
- Developing a third research manuscript investigating how high-frequency order-flow memory relates to medium-frequency persistence and regime structure in equity returns

Research and Teaching Assistant with Prof. Ren Raw Chen

January 2025 – May 2026

- Modeled 13F data (4,634 funds, 4,824 assets) as a bipartite network and constructed graph embeddings using node2vec
- Measured latent fund similarity to study liquidity clustering and potential fire-sale contagion channels among funds
- Teaching assistant for 2 classes: Financial Theory and AI in Asset Management

Capstone Research sponsored by AutoCase Economic Advisory

September 2025 – December 2025

- Marine WTP analysis for a Canadian estuary: used DAG-derived causal features with Ordered Probit Selection model to address biased demographic survey data and model latent attitudinal variable, to improve R^2 from 0.08 to 0.23

EXPERIENCE

Rebellion Research*Quant Research Intern (Python)*

Remote, NY

June 2025 – September 2025

- Studied 13F data to analyze common holdings of portfolio managers to build internal report and trading strategy
- Authored and published an interview with Deutsche Bank's Global Quant Head on the evolution of the QIS space

Independent Quantitative Research and Pre-Grad Study

May 2023 – May 2024

- Self-study in Real Analysis, Probability Theory, Linear Algebra, Stochastic Processes, Asymptotic Theory
- Evaluated retail-sales data as a leading indicator of Walmart revenue using a walk-forward forecasting framework; achieved a 14% reduction in out-of-sample RMSE versus a naive seasonal baseline
- Code and memo: [Signal Incrementality Analysis for Revenue Forecasting](#)
- Implemented risk targeting strategies from Carver's textbook; Strategy 5 attained Sharpe ratio of **1.2** after 10 bps tc over 2005–2025 (walk-forward, 35 diversified futures; consistent on crypto portfolio)
- Code and presentation: [Carver Trend-Following Implementation](#)

FlexTrade*Associate Software Developer (C++)*

Pune, India

February 2023 – May 2023

- Integrated inventory-management and fair-value pricing engines in C++, coordinating order cancellation, re-quoting, and order-splitting logic as market prices deviated from theoretical value
- Optimized latency and memory in a market-making system, understanding inventory management and order-routing workflows

Interactive Brokers Software Services*Junior Software Engineer (Java)*

Mumbai, India

August 2022 – December 2022

- Maintained and extended trade-matching engine functionality in Java through bug fixes, edge-case handling
- Contributed to a multithreaded platform designed to support heterogeneous report formats through a scalable architecture

ADDITIONAL

Programming: Python, C++, Java; NumPy, Pandas, SciPy, statsmodels, scikit-learn, TensorFlow**Methods:** Time-Series Econ., Stochastic Calc., Statistical Arb., Causal Inference, Monte Carlo Simulation, Computer Vision**Tools:** Git, Linux/Bash, SQL, Multithreading, OOP, Data Structures and Algorithms**Other Interests:** Chess (FIDE 1500); Gym; Reading; UCMAS(mental math); MUNs